VICTORY 2027

Sense Organs

LECTURE-1 BIOLOGY

Upasana ma’am



Topics To Be Covered

1. Sense organs
2. External and internal structure of eye



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RATTON KO KARO BYE BYE !!

EASY CONCEPTS SE KARO HELLO HI !!

With – Uppu Ma'am



SENSE ORGANS

- The sense organs enable us to be aware of the conditions of our external as well as internal environment.



Sight Smell Touch Hearing Taste

SENSE ORGANS

- The major sense organs in our body are the eyes, ears, tongue, nose and skin which are sensitive to light, sound, chemical substances (responsible for taste and smell) and touch respectively.

temperature



Sight



Smell



Touch



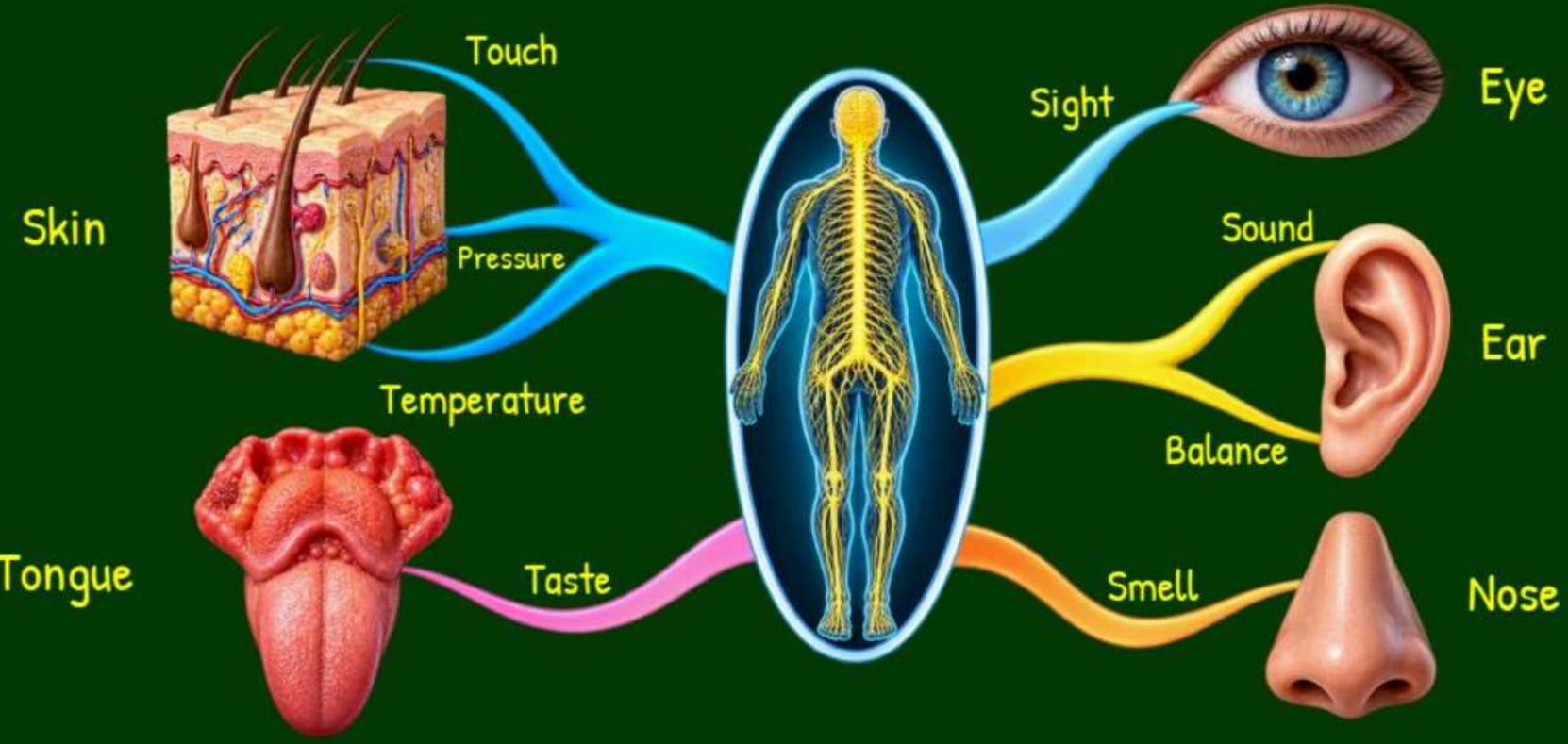
Hearing



Taste

SENSE ORGANS

- There are also the senses of balance, body movements, hunger, thirst and pain, etc.





SENSE ORGANS

- The actual sensation is perceived by the sensory cells located in these organs - such cells are categorised as receptors.

RECEPTORS

- Receptor is any specialised tissue or cell sensitive to a specific stimulus.

Photo → light

(i) Photo-receptors - are rods and cones of the retina of eye.
These are sensitive to light and are responsible for vision.

→ Sound

(ii) Phono-receptors - are receptors present in the inner ear.
These are sensitive to sound and are responsible for hearing and balance.

(Taste)
Gustatory Receptors
Olfactory

(iii) **Chemoreceptors** - are receptors of the tongue and nose. They are sensitive to chemical substances and are responsible for taste and smell.

Temperature

(iv) **Thermoreceptors** - are temperature-sensing receptors present in the skin. They are sensitive to relative changes in the temperatures and are responsible for our body feeling hot or cold.

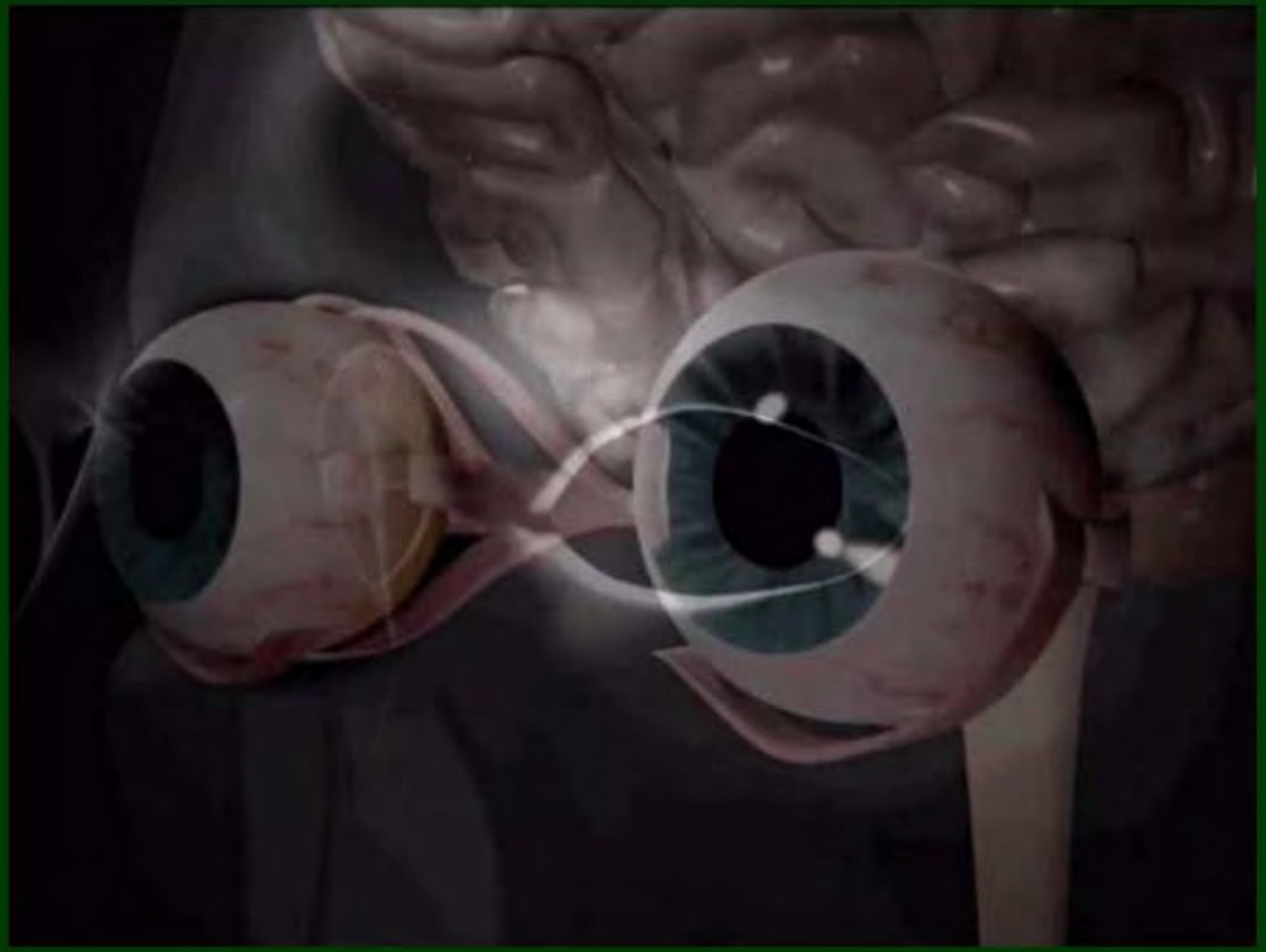
(v) **Mechanoreceptors** - are receptors present in the skin. They are sensitive to different mechanical stimuli such as touch, pressure, vibrations, etc.



ORBITS

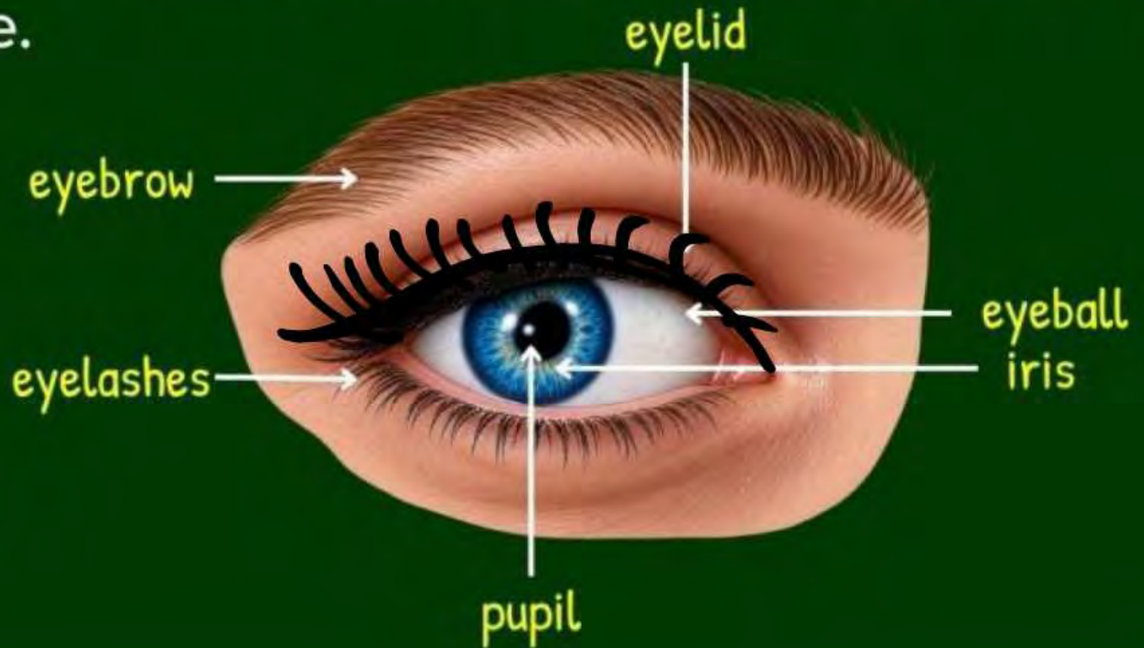
- The two eyes are located in deep sockets or orbits on the front side of the head.
- Each eye is in the form of a ball and can be rotated with the help of six muscles.





EYELIDS

- The upper (movable) and the lower eyelids protect the outer (front) surface of the eyes and can shut out light.
- Each eyelid carries outwardly curved eyelashes which prevent falling of larger particles into the eye.

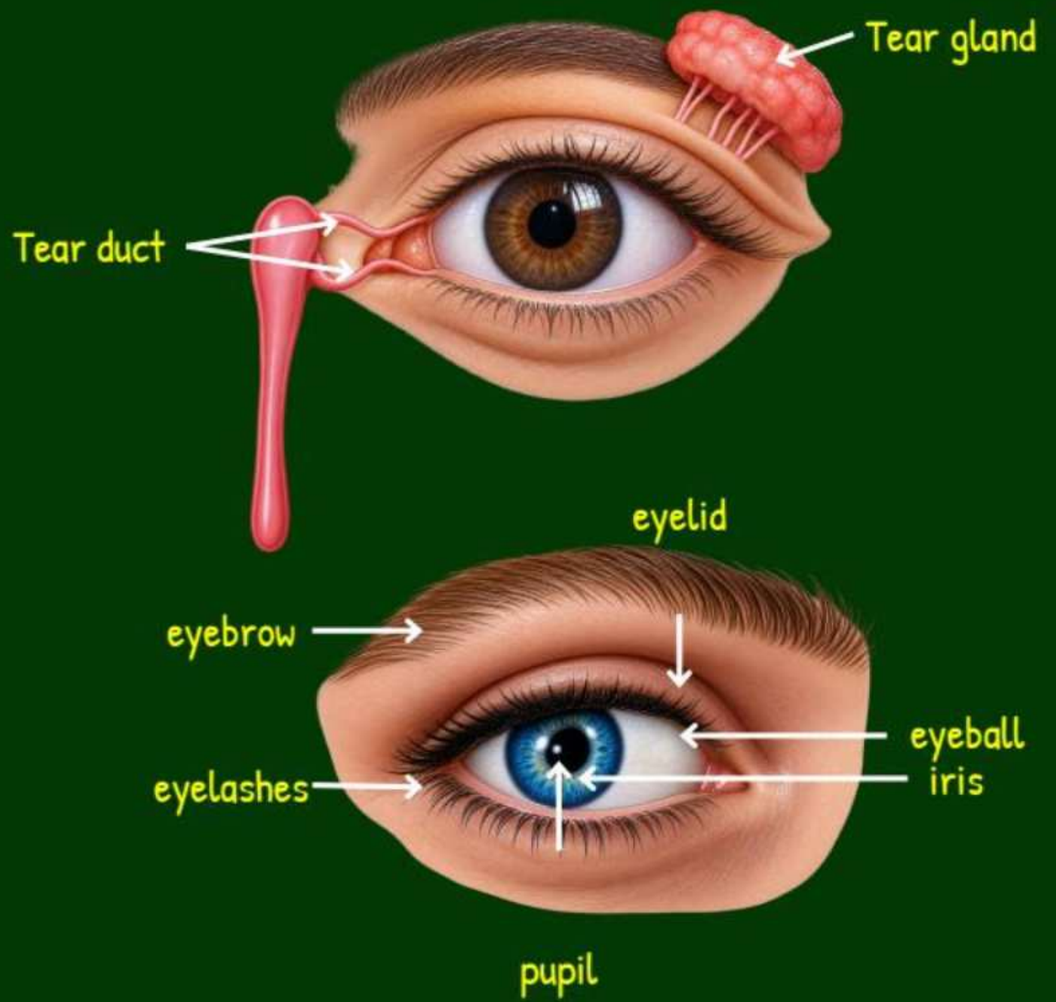



Sclera
Cornea
Pupil
Iris
Lens
Ciliary muscles
Vitreous
Retina
(cones &rods)
Optic Nerve

EYEBROWS

- Not a part of the eye, they prevent the rain drops or the trickling perspiration from getting into the eyes.

- Tear glands (lacrimal glands): They are located at the upper sideward portion of the orbit. Six to twelve ducts of the gland pour the secretion over the front surface.

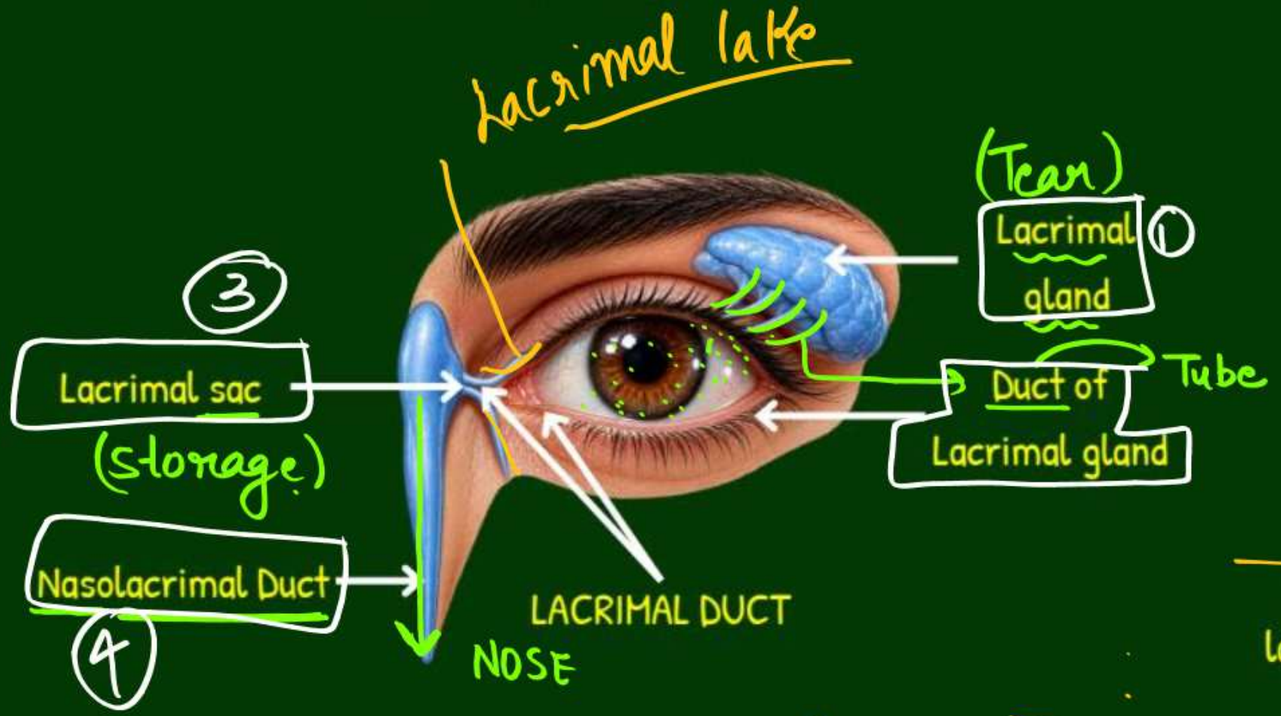


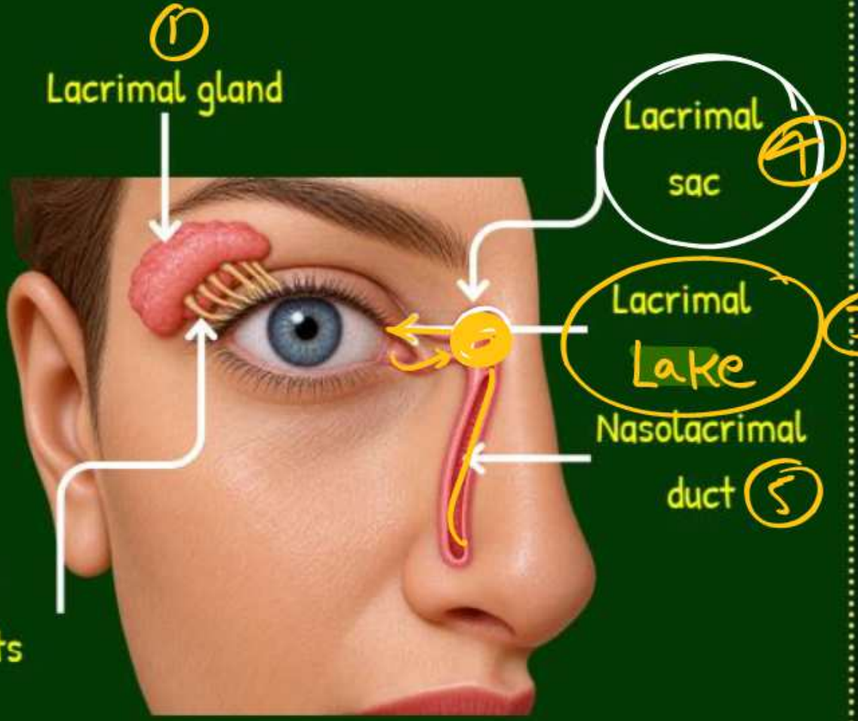
EYE

- The movements of the eyelids (blinking) spread the liquid which mainly serves as a lubricant.
- The tears also keep the front surface of eye clean by washing away dust particle.
- They also have an antiseptic property to the enzyme lysozyme which kills the germs.



- Lacrimal (Lear) apparatus of the human eye





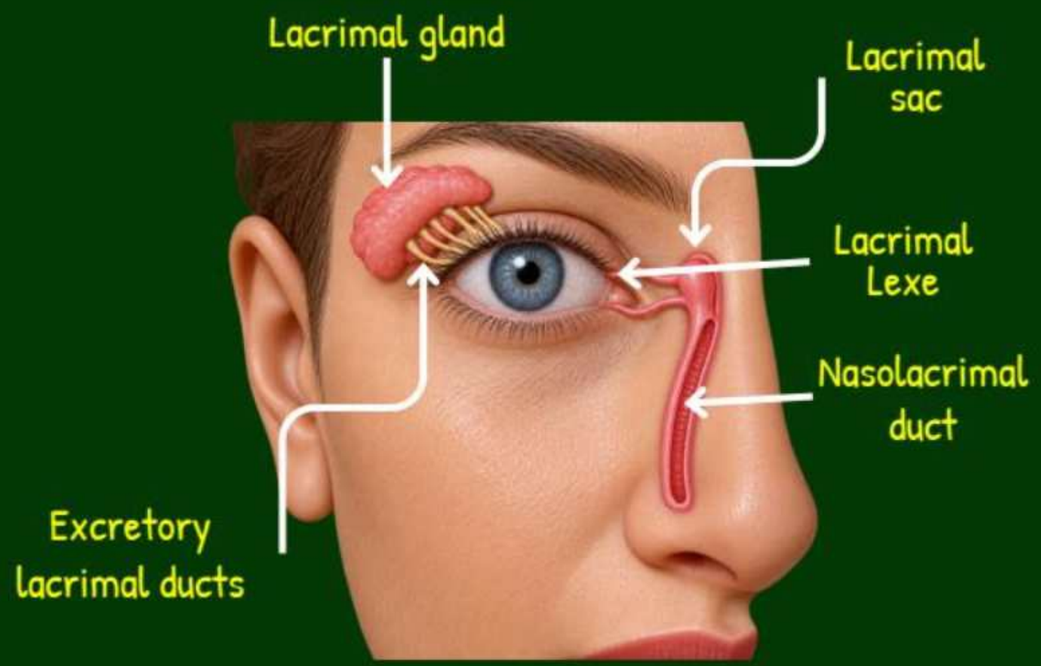
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Excretory lacrimal ducts

Lacrimal (Lear) apparatus of the human eye

TEAR DUCTS

L. gland → 6-12 L duct → L. sac → N.L. duct

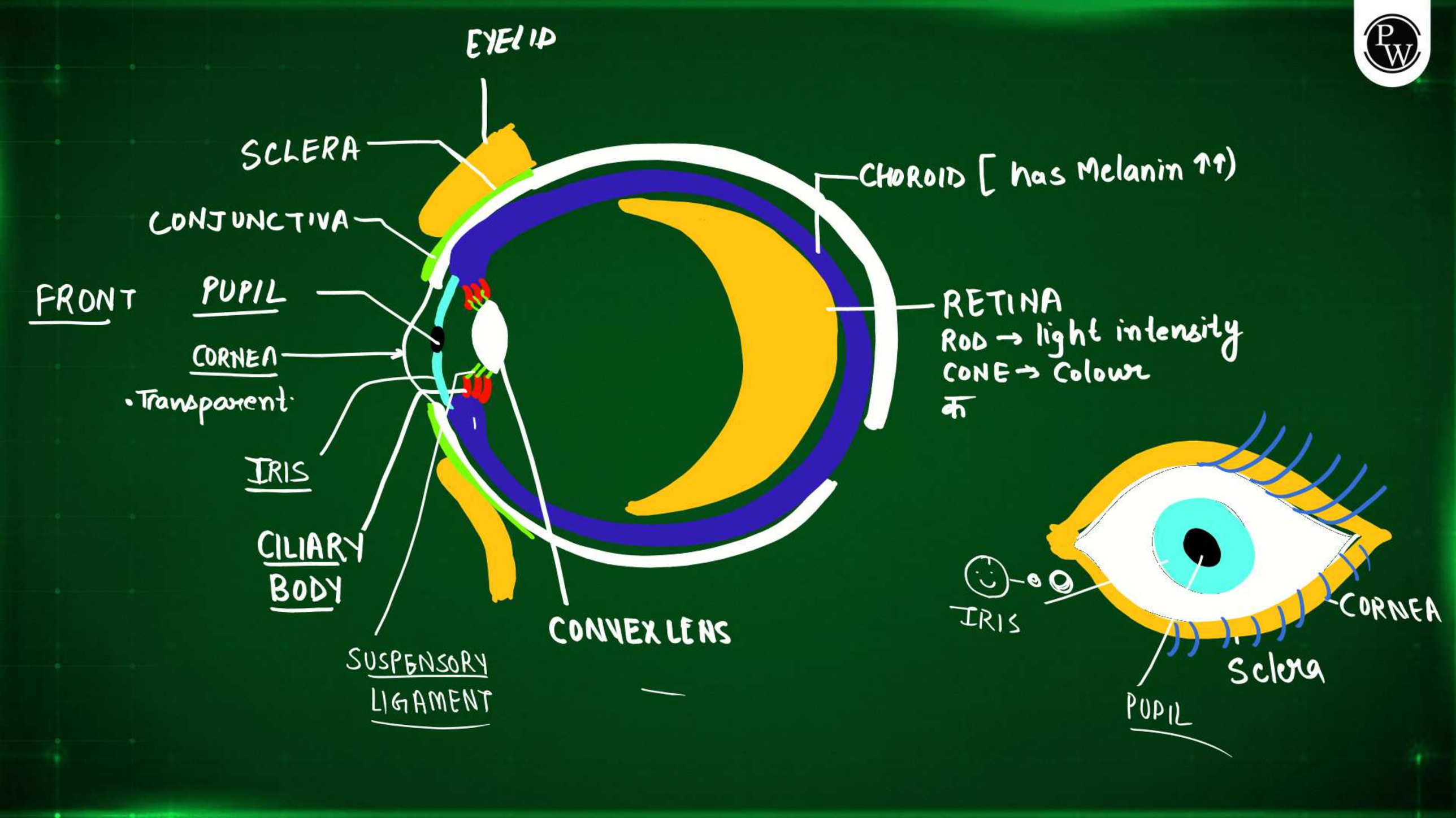
- These ducts drain off the liquid into a sac lying at the inner angle of the eye.
- A nasolacrimal duct conducts the secretion into the nasal cavity.



FUNCTIONS OF TEARS

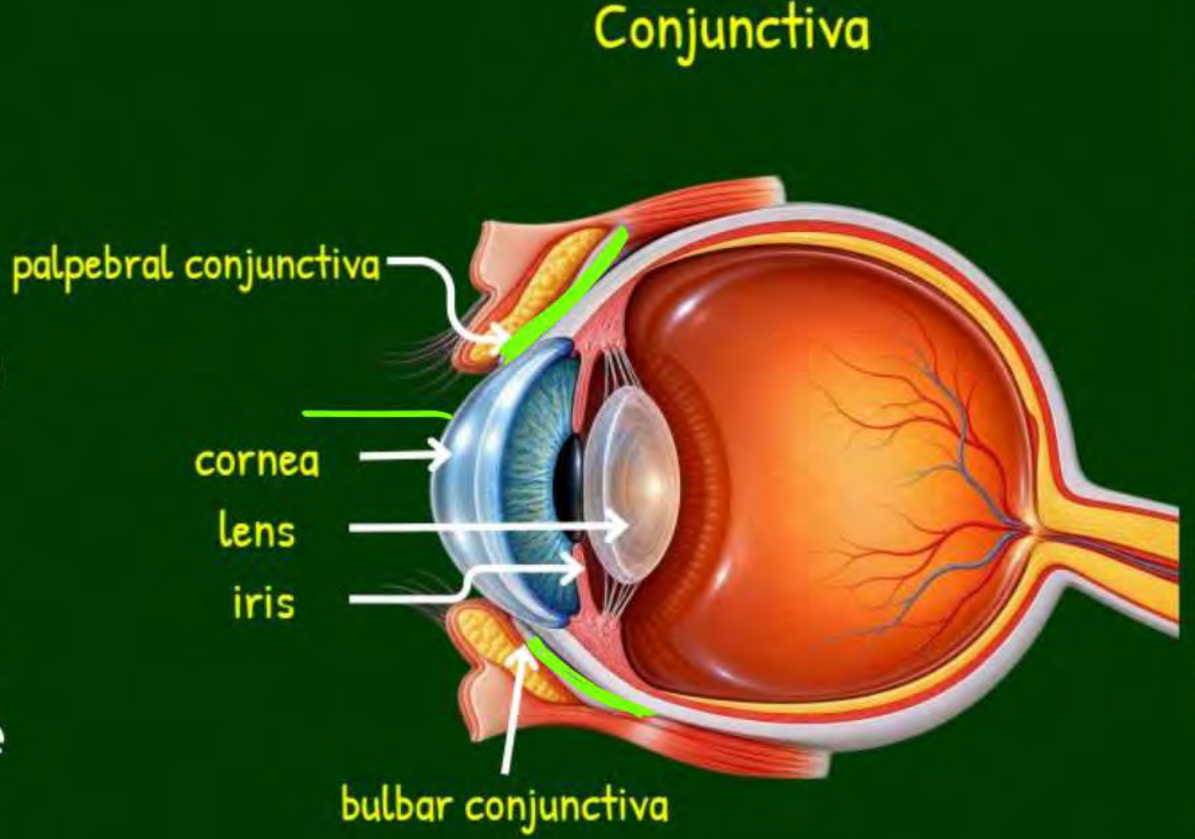
1. Lubricate the surface of the eye.
2. Wash away dust particles.
3. Help in killing germs.
4. Communicate emotions.

lysozyme.


EYELID
SCLERA
CONJUNCTIVA
FRONT
PUPIL
CORNEA
• Transparent
IRIS
CILIARY BODY
SUSPENSORY LIGAMENT
CONVEX LENS
CHOROID [has Melanin ↑↑)
RETINA
ROD → light intensity
CONE → Colour
IRIS
CORNEA
Sclera
PUPIL

CONJUNCTIVA

- It is a thin membrane covering the entire front part of the eye.
- It is continuous with the inner lining of the eyelids.
- Over the cornea, it is reduced to a single layer of transparent epithelium.
- The bony orbit socket, eyebrows, tear glands and conjunctiva serve for eye protection.



QUESTIONS

The layer in the eye where sensory cells (rods and cones) are located:

A Conjunctiva

B Cornea

C Choroid

D Retina

QUESTIONS

An aperture that through which the light enters?

A. Blind spot

B. Pupil

C. Yellow spot

D. Iris

QUESTIONS

Tears have an antiseptic property due to the presence of:

A Lysosome

B Aqueous humour

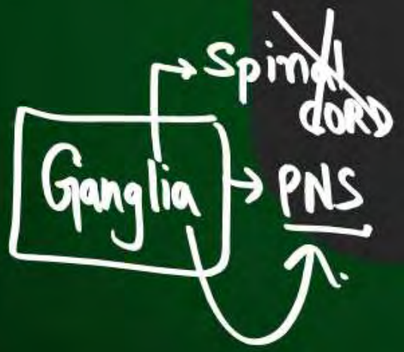
C Lysozyme

D Vitreous humour

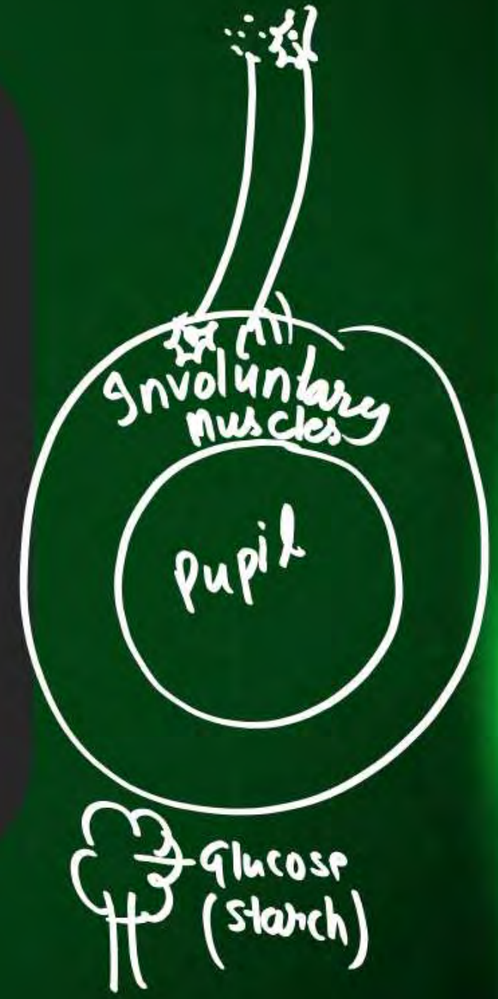




Thank You



Alfaaz



(Nuclei) → false